

Introduction to “Designing for Real People” 2nd Year Group Projects

The Imperial-RCA DRP Coordination Team:

Mark Wheelhouse mark.wheelhouse@imperial.ac.uk

Carolyn Runcie carolyn.runcie@rca.ac.uk

with Software Engineering input from

Rob Chatley r.chatley@imperial.ac.uk

Ivan Procaccini ivan.procaccini@imperial.ac.uk

Summer Term Outline

Spring Week 11: HCD Lectures – Introduction to HCD, Discover & Define
Application Framework Tech Talks (pre-recorded)
Industrial Project Pitches (shared on EdStem)
HCD Workshop (Friday)

Summer Weeks 1 - 3: Exams

Summer Week 4: DRP “Inception” Week
HCD Lectures – Storytelling, Testing, Prototyping & User Experience

Summer Week 5: DRP Project Iteration 1 - Walking Skeleton
HCD Lectures – Service Systems
Introduction to Law for Computer Science

Summer Week 6: DRP Project Iteration 2 - Thin Slicing
Introduction to Law for Computer Science - Case Study TRA

Summer Week 7: DRP Project Iteration 3 - Quantitative Evaluation

Summer Week 8: DRP Final Demonstrations & Presentations
DRP Project Documentation Deadline

Designing for Real People Project

During the “Designing for Real People” (DRP) Project you are required to design and implement a solution to a **real-world problem** identified by teams who understand a chosen context that involves real people.

Understanding this context helps you to identify opportunities to address specific problems and should include a digital touchpoint in the form of a **multi-user** web or mobile application.

Your digital touchpoint should involve **real-time** user interactions and must use a **database** to track some persistent internal state.
(Further details can be found in the DRP project specification)

Focus on something of interest to you and try to think of something original!

Make sure you have access to representative users (not yourself!)

Human Centred Design (HCD)

Alongside some of the more technical talks, a range of design lectures have been created to help you understand the principles, techniques and best practices of Human Centred Design. In particular, how to identify a good **real-world** problem and design an appropriate solution to it.

The information you learn in these sessions should feed ***directly*** into your DRP projects, as well as all future project work during your degree and beyond.

Aspects of HCD are fully integrated into the Milestone Assessments of your DRP project and will be assessed by both Imperial DoC and Royal College of Art tutors, recognising the importance of understanding the behaviours and needs of people in their daily lives.

(i.e. no additional submissions are required)

Designing for Real People Project

You will work through the project in a series of **iterations** supported by targeted Software Engineering and Human Centred Design Lectures, as well as Lab Clinic Hours.

Assessment is **continuous** throughout these iterations:

Milestone Assessments – 20% (5% / 5% / 5% / 5% over 4 weeks)

Law Case Study TRA – 10%

Final Demonstration and Presentation – 50%

Project Documentation Producibles – 20%

Note: whilst we have given you the project brief and specification early, we do **not** expect you to start work on your projects during your revision and exam periods.

Application Framework Tech Talks

Over the years, we have made some recordings of a number of invited talks introducing some popular web-development frameworks.

The frameworks we intend to provide material for are:

- **JavaScript** and **React**
- **Scala JS**
- **Python** and **Django**
- **Python** and **Flask**
- **iOS** Development with **Swift**

These will be released on Panopto (with the permission of the guest speakers). We may also invite additional guest speakers to prepare similar recordings for other popular web-development frameworks during the Summer term.

Industrial Project Pitches

We have organised a series of ***project idea pitches*** from the Department's industrial contacts. These “approved” pitches provide possible jumping off points for a DRP project to work with an external client.

We are gathering these proposals together and will share them with you online in the coming days.

Groups are ***not*** required to pick an industrial proposal to work on.

Intro. to Law for Computer Scientists

A short sub-module aiming to give you an overview of the legal aspects of working in the digital era, including:

- **copyright and licensing**
- **intellectual property**
- **contractual obligations**
- **data protection**

The information you learn should feed ***directly*** into your DRP projects, as well as all future project work during your degree.

Assessment is over one submission:

Law Case Study TRA – 100%

(A component of the DRP project documentation assessment is also directly related to this sub-module.)

Important Dates

19th March: DRP group organisation (max 4 students)

20th March: HCD Workshop (practical activities with tutor support)

18th – 21st May: DRP Clinic Hours (in Labs)

22nd May: DRP Milestone Assessment 1 (Elevator-Pitch)

26th – 27th May: DRP Clinic Hours (in Labs)

29th May: DRP Milestone Assessment 2 (Progress Meeting)

1st – 2nd June: DRP Clinic Hours (in Labs)

2nd June: Law Case-Study TRA (10.00 - 12.00)

5th June: DRP Milestone Assessment 3 (Progress Meeting)

8th – 9th June: DRP Clinic Hours (in Labs)

12th June: DRP Milestone Assessment 4 (Progress Meeting)

15th June: DRP Clinic Hours (in Labs)

16th – 17th June: DRP Final Demonstrations and Presentations

19th June: DRP project documentation producibles deadline (19.00)

Preparing for the DRP Workshop (Friday 20th March)

1. Form your DRP team (before Friday)
2. Choose a brief that interests you most (next slide)
3. Carry out some desktop research to identify a specific challenge or opportunity. Bring facts and statistics with sources, as well as an idea on who is the target audience (who's most affected in your chosen context)

In your teams, choose your favourite topic from the list below:

1. **What a waste!:** 215 million tonnes of waste is produced in the U.K each year [Waste Direct 2024].
2. **Roundabout:** London's workers spend an average of 107 hours commuting annually, more if they have disabilities [Gov.uk, 2017].
3. **Ker-ching:** Nearly 60% of UK parents have to help their adult kids (ages 18 to 34) with financial support. [[pewresearch.org](https://www.pewresearch.org), 2024].

Discuss some key assumptions, areas of interest you're already aware of through reading the news, etc...

... and finally ...

Good Luck
with
Your Exams!